

Software Requirements Specification

LifeLink: An Intelligent Blood & Organ Coordination Network

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Revision History

Name	Date	Reason For Changes	Version
Shivanshu Khode, Sarthak Vyas, Aryan Gupta	19-Sep-2026	Initial SRS drafted from approved project synopsis and finalized architecture context.	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for Version 1.0 of LifeLink: An Intelligent Blood & Organ Coordination Network. LifeLink is a web-based platform connecting hospitals, blood banks, organ centres, individual users, donors, recipients, and administrators for structured resource discovery, requesting, and coordination.

This SRS covers the complete LifeLink system, including blood and organ coordination, map-based discovery, emergency notifications, real-time workflow tracking, predictive analytics, RAG-based assistance, document management, access control, and audit logging. It serves as the reference for system design, implementation, testing, and evaluation.

1.2 Document Conventions

This document follows a consistent naming and identification convention for requirements, system features, and other technical elements to improve readability, traceability, and ease of reference.

Requirement Identification:

Functional requirements are identified using the following format:

LL-FR-XXX-NNN

where:

- LL → LifeLink
- FR → Functional Requirement
- XXX → Abbreviation representing the relevant system feature or module
- NNN → Sequential number of the requirement within that feature

For example, LL-FR-INV-001 represents the first functional requirement under the Blood Inventory Management feature.

Common feature abbreviations include:

- AUTH → User Authentication and Role Management
- INST → Institution and Profile Management
- INV → Blood Inventory Management
- BREQ → Blood Request and Matching
- MAP → Map-Based Discovery

- EMG → Emergency Coordination
- ORG → Organ Coordination
- RTN → Real-Time Coordination and Notifications
- RAG → RAG-Based Intelligent Assistance
- PRED → Predictive Analytics
- DOC → Document and Record Management
- ADM → Administration and Audit
- Other requirement categories use similar identifiers:
- PERF-XXX → Performance Requirement
- SAFE-XXX → Safety Requirement
- SEC-XXX → Security Requirement
- QUAL-XXX → Software Quality Attribute Requirement
- BR-XXX → Business Rule

The terms “shall” “should”, and “may” are used consistently. “Shall” indicates a mandatory system requirement, “should” indicates a recommended behaviour or quality, and “may” indicates an optional capability.

Priority is expressed as High, Medium, or Low where applicable. Requirements and system features are numbered according to their corresponding section and subsection in the document.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the development team, project supervisor and evaluation panel, testers, and future contributors.

Readers should begin with Section 1 and Section 2 to understand the system, actors, scope, and architecture. Developers should then refer to Section 3 for interfaces and Section 4 for functional requirements. Section 5 defines requirements applicable across the system, while Section 6 and the appendices provide additional requirements, definitions, analysis models, and TBD items.

1.4 Product Scope

LifeLink is a regional, web-based healthcare coordination platform designed to reduce the delay and manual effort involved in locating and coordinating blood and organ resources through a shared system across registered institutions.

The platform provides:

- Centralised and role-controlled visibility of blood inventory and organ coordination records.

- Blood and organ request, matching, acceptance, rejection, and workflow management.
- Emergency broadcasting, notifications, and real-time updates.
- Map-based discovery of nearby hospitals, blood banks, and organ centres using Leaflet and OpenStreetMap.
- Rule-based organ compatibility filtering as system-level decision support.
- Blood-demand forecasting and shortage-risk classification.
- RAG-based assistance using approved documents, SOPs, policies, and FAQs.
- Role-based access control, privacy protection, document management, and audit logging.

LifeLink does not replace or integrate with systems such as e-RaktKosh or NOTTO's e-Pratyaropan portal and does not perform autonomous clinical decision-making, transfusion approval, or organ allocation.

1.5 References

Ref	Title / Source
[1]	e-RaktKosh: Centralised Blood Bank Management System, C-DAC / National Health Mission, Government of India — www.eraktkosh.in
[2]	National Organ and Tissue Transplant Organisation (NOTTO) — e-Pratyaropan National Organ Transplant Portal, Ministry of Health and Family Welfare, Government of India, 2026 — notto.mohfw.gov.in
[3]	United Network for Organ Sharing (UNOS) — UNet Organ Matching Technology — www.unos.org
[4]	Indian Red Cross Society, “eBloodServices / Blood Bank Services”
[5]	BloodConnect Foundation, “BloodConnect: Blood Donation and Donor Mobilisation Network”, India
[6]	P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, vol. 33, 2020, pp. 9459–9474
[7]	Google, “Gemini API Documentation – Gemini 3.1 Flash-Lite and Gemini Embedding 2,” Google AI for Developers, 2026
[8]	LifeLink Project Synopsis, Department of CSE, IET DAVV, 2026
[9]	S. Kaur, P. Singh, and A. Sharma, "Blood demand forecasting using hybrid ARIMA-LSTM model: A case study in Indian blood banks," Computers in Biology and Medicine, vol. 154, 2023, p. 106506.
[10]	Karl E. Wiegers, Software Requirements Specification template, 1999 (base document format used for this SRS)

2. Overall Description

2.1 Product Perspective

LifeLink is a web-based healthcare coordination platform designed to provide a centralized environment for coordinating blood- and organ-related requirements among hospitals, blood centres, organ centres, and users. It integrates institutional resource management, requirement coordination, map-based discovery, emergency communication, real-time status updates, notifications, and intelligent assistance within a unified platform.

The platform is intended for regional/city-level coordination and facilitates information exchange between participating healthcare institutions and users. Operational information such as blood inventory, requests, donor and recipient records, and workflow statuses is managed through the application and database services, while map-based functionality supports location-aware discovery of nearby healthcare facilities.

LifeLink incorporates an intelligence layer comprising RAG, a large language model, and predictive analytics to support information retrieval, natural-language interaction, and blood-demand analysis. These components provide assistance and decision support rather than autonomous clinical or regulatory decision-making.

LifeLink is intended to complement existing healthcare coordination processes and does not replace regulated healthcare systems, professional clinical decisions, or official organ-allocation authorities.

2.2 Product Functions

LifeLink provides integrated functions for coordinating blood- and organ-related healthcare requirements among authorized users and participating institutions. The major functions include:

- **User and Access Management:** Supports registration, authentication, profiles, role-based access, and authorized account management.
- **Institution Management:** Enables hospitals, blood centres, and organ centres to maintain profiles and relevant operational information.
- **Blood Inventory Management:** Enables authorized institutions to maintain and update blood-resource availability information.
- **Blood Request and Coordination:** Supports creation of blood requirements, resource discovery, coordination, and request-status tracking.
- **Map-Based Discovery:** Enables location-based discovery of nearby hospitals, blood centres, and organ centres.
- **Emergency Coordination:** Supports prioritized requirements, nearby resource discovery, emergency notifications, broadcasting, and response tracking.
- **Organ Coordination:** Supports donor and organ records, recipient requirements, matching and offer coordination, procurement, transport/status tracking, and related records.
- **Real-Time Coordination and Notifications:** Provides real-time workflow updates and notifications to relevant users and institutions.

- **RAG-Based Intelligent Assistance:** Provides context-aware assistance using approved knowledge sources and retrieval-augmented generation.
- **Predictive Analytics:** Supports blood-demand forecasting and shortage-risk classification for resource planning.
- **Document and Record Management:** Supports management of documents and records associated with users, institutions, requests, and workflows.
- **Administration and Audit:** Provides administrative oversight, access management, monitoring, and audit capabilities.

2.3 User Classes and Characteristics

LifeLink supports multiple user classes with different responsibilities and access privileges. Role-based access control is used to provide users with functions and information appropriate to their assigned roles. Users are expected to have basic familiarity with web-based applications. Role-specific guidance is provided for functions requiring additional operational knowledge.

User Class	Characteristics
Hospital	Hospital users manage hospital information and participate in blood, emergency, and other applicable healthcare coordination workflows. Hospital profiles may display associated blood centres or organ centres where such relationships are recorded.
Blood Bank	Blood centre users maintain blood inventory information, respond to blood requirements, and participate in regular or emergency coordination. Blood centres are independent entities and may optionally be associated with a hospital or other institution.
Organ Centre	Organ centre users manage authorized organ-coordination workflows, including donor and organ records, recipient requirements, matching, offers, procurement, and transport-related tracking. Organ centres are independent entities and may optionally be associated with a hospital or other institution.
User (Donor/Recipient)	The User represents an individual interacting with LifeLink. Depending on authorization and workflow context, a user may request blood, participate in donor or recipient workflows, discover healthcare facilities, track requests, receive notifications, and use intelligent assistance. Donor and recipient roles are represented as roles or workflow contexts within the User entity rather than separate top-level actors.
Administrator	Administrators perform system-level oversight, including account management, access control, monitoring, and audit-related activities.

2.4 Operating Environment

LifeLink is designed as a web-based application accessible through modern web browsers on desktop and mobile computing devices.

- The frontend uses Next.js, React, TypeScript, and Tailwind CSS, while backend services use Node.js, Express.js, and TypeScript.
- PostgreSQL with Prisma provides the primary operational data layer.
- Redis and Socket.IO support caching, publish/subscribe operations, and real-time communication.

- Map-based discovery uses Leaflet with OpenStreetMap-based services, while object storage and vector storage support document management and the RAG knowledge layer.
- Predictive analytics is provided through a Python and scikit-learn service, while Gemini Embedding 2 and Gemini 3.1 Flash-Lite support the embedding and language-model components of the intelligence layer.
- The system is developed and deployed using Docker, with Git used for source-code version control.
- Network connectivity is required for communication between clients, backend services, databases, and applicable external services.

2.5 Design and Implementation Constraints

LifeLink shall be implemented using the finalized technology stack and architecture defined for the system, including the specified frontend, backend, database, real-time, mapping, storage, and intelligence technologies.

- The system shall use role-based access control to separate functionality and information according to user responsibilities. Protected operations shall be performed through authorized backend services rather than relying solely on client-side controls.
- Operational and transactional information shall be maintained through the structured operational database and authorized backend services.
- The RAG knowledge store shall remain separate from the operational data source and shall not be treated as the source of truth for live information such as current inventory or request status.
- The intelligence layer shall remain within a decision-support role. AI and predictive components shall not independently perform clinical, medical, or regulatory decisions.
- The system shall maintain appropriate privacy, security, and audit controls for sensitive information and important system activities.
- The system is intended for regional/city-level coordination, and its functionality depends on the quality and availability of participating data and required external services. It shall not be represented as a replacement for regulated healthcare coordination or official organ-allocation systems.
- Hospital, blood centre, and organ centre records shall remain separate entities, while optional institutional associations may be maintained between them.

2.6 User Documentation

LifeLink shall provide documentation to help users understand and operate the system according to their roles. Documentation shall cover major functions, account and access procedures, and relevant coordination workflows.

Role-specific guidance shall be provided for hospitals, blood centres, organ centres, general users, and administrators, including applicable workflows such as blood coordination, emergency coordination, map-based discovery, organ coordination, notifications, and status tracking.

Documentation shall also describe the purpose and limitations of RAG-based and AI-assisted functionality, including the distinction between system assistance and professional medical or regulatory decision-making.

Administrative guidance shall cover relevant account management, access control, monitoring, and audit functions. Documentation may be provided through a user manual, in-system to help content, or other appropriate system documentation.

2.7 Assumptions and Dependencies

- LifeLink assumes that participating hospitals, blood centres, organ centres, and users provide sufficiently accurate and current information. The effectiveness of availability and coordination functions therefore depends on the quality and timeliness of data maintained by authorized users and institutions.
- The system assumes that users have appropriate authorization for the information and functions they access, along with a compatible web browser and reliable internet connectivity.
- LifeLink depends on the availability of its core infrastructure, including backend services, PostgreSQL, Redis, vector storage, and object storage, as well as external services used for mapping, AI-assisted functionality, and other integrated capabilities.
- RAG functionality depends on the quality, completeness, and maintenance of its approved knowledge sources. Predictive analytics similarly depends on the availability of sufficiently relevant and authorized historical data.
- Temporary unavailability of an external service may affect the corresponding functionality without necessarily affecting unrelated core services.

3. External Interface Requirements

3.1 User Interfaces

LifeLink provides a responsive, role-based web interface using Next.js, React, TypeScript, and Tailwind CSS. Each authenticated role (Hospital, Blood Bank, Organ Centre, User, Administrator) receives only its authorized functions.

- Dashboards — Display relevant inventory, requests, notifications, and workflow status.
- Map-Based Discovery — Leaflet/OpenStreetMap map for nearby hospitals, blood banks, and organ centres with distance/radius filtering.
- Blood & Organ Search — Structured forms for blood group/component, quantity, priority, location, and radius, with relevant availability and distance information.
- Reports & Insights — Displays inventory trends, request outcomes, demand forecasts, and shortage-risk indicators.
- AI Assistant — Provides RAG-grounded natural-language assistance and workflow guidance.

Common conventions include consistent navigation, action buttons, validation feedback, notifications, and confirmation dialogs for significant state changes. Errors shall be clear and shall not expose sensitive system details. The interface shall support desktop, laptop, tablet, and smartphone screens. No application-specific keyboard shortcuts are required.

Interfaces are required for authentication, profiles, coordination workflows, maps, reports, notifications, documents, AI assistance, and administration. Detailed mockups are maintained separately in a User Interface Specification.

3.2 Hardware Interfaces

LifeLink requires no specialised hardware in the current version. It operates on standard desktop, laptop, tablet, or smartphone devices through a supported web browser and internet connection. Input is provided through keyboard, mouse, or touch interfaces. Future interfaces such as barcode/RFID scanners, biometric devices, or dedicated GPS hardware shall be specified separately if introduced.

3.3 Software Interfaces

Component	Purpose / Data Exchanged
Next.js / React / TypeScript / Tailwind CSS (Frontend)	Frontend; sends user input and receives REST responses and Socket.IO events.
Node.js + Express.js (Application Layer)	handles authentication/RBAC, coordination, notifications, geospatial services, and RAG orchestration.
PostgreSQL via Prisma ORM	Stores operational users, institutions, inventory, requests, notifications, documents, predictions, and audit data; operational source of truth.
Vector Store	Stores document chunks and embeddings for RAG retrieval.
Redis + Socket.IO	Provides caching, pub/sub, rate limiting, and real-time updates.
Leaflet + OpenStreetMap	Provides map rendering and location-based discovery.

Component	Purpose / Data Exchanged
Python + scikit-learn (Predictive Analytics Service)	Produces demand forecasts and shortage-risk classifications.
Gemini 3.1 Flash-Lite / Gemini Embedding 2	Provides RAG-based generation and document/query embeddings.
Object Storage	Stores supporting files; PostgreSQL stores metadata/references.
Docker / Git	Support deployment and version control.

Component versions not finalized are [TBD]. Shared data includes users, institutions, inventory, requests, workflow states, notifications, documents, predictions, RAG data, and audit records. PostgreSQL remains the authoritative persistent store; Redis supports temporary and real-time functions. Detailed API specifications may be maintained separately.

3.4 Communications Interfaces

LifeLink uses HTTPS with JSON-formatted REST requests/responses for client-server communication. Real-time inventory, request-status, notification, and emergency events use Socket.IO, with Redis pub/sub where required.

The Notification Service supports e-mail, SMS, and in-app notifications; final external providers are [TBD]. Gemini API communication occurs over HTTPS.

Client-server and external API communication carrying LifeLink data shall use HTTPS/TLS. No fixed bandwidth requirement is imposed; communication shall meet the performance targets in Section 5.1. PostgreSQL remains the authoritative state source, and communication failures shall not be reported as successful state-changing operations.

4. System Features

This section groups the LifeLink functional requirements into twelve user-observable system features. Each feature states its description and priority, a representative stimulus/response sequence, and its uniquely identified functional requirements, written in testable “shall” language. Requirements describe externally observable system behavior; they are not simply a restatement of the architecture diagram in Section 2.1.

4.1 User Authentication and Role Management

4.1.1 Description and Priority

This feature provides secure user registration, authentication, account management, and role-based access to LifeLink. It ensures that users can access only the functions and information permitted by their assigned roles.

Priority: High

4.1.2 Stimulus/Response Sequences

- A user registers or logs in → the system validates the provided credentials → the system authenticates the user → the applicable role and permissions are identified → the user is granted access to the corresponding dashboard and functions.
- When an administrator changes a user's role or access status → the system validates the authorization → updates the account permissions → applies the changes to subsequent protected operations.

4.1.3 Functional Requirements

LL-FR-AUTH-001: The system shall allow users to register accounts using required identity and contact information.

LL-FR-AUTH-002: The system shall authenticate users using valid credentials.

LL-FR-AUTH-003: The system shall reject invalid or unauthorized login attempts.

LL-FR-AUTH-004: The system shall provide password reset and account-recovery functionality.

LL-FR-AUTH-005: The system shall maintain user profiles containing authorized personal and contact information.

LL-FR-AUTH-006: The system shall assign and maintain roles for authorized users.

LL-FR-AUTH-007: The system shall enforce role-based access control for protected functions and information.

LL-FR-AUTH-008: The system shall allow authorized users to update permitted profile information.

LL-FR-AUTH-009: The system shall allow authorized administrators to activate, deactivate, or restrict user accounts.

LL-FR-AUTH-010: The system shall terminate or invalidate authenticated sessions according to defined session rules.

LL-FR-AUTH-011: The system shall prevent unauthorized users from accessing protected resources through direct requests.

LL-FR-AUTH-012: The system shall maintain appropriate records of significant authentication and access-control activities.

LL-FR-AUTH-013: The system shall apply authorization checks through backend services for protected operations.

4.2 Institution and Profile Management

4.2.1 Description and Priority

This feature manages profiles and relevant information for hospitals, blood centres, and organ centres participating in LifeLink. These entities remain independently represented while optional institutional associations may be maintained.

Priority: High

4.2.2 Stimulus/Response Sequences

- An authorized institution user creates or updates an institution profile → the system validates the information → stores the changes → makes the permitted information available for discovery and coordination.
- When an association between a centre and an institution is created or changed → the system validates the relationship → updates the relevant records → reflects the association in authorized views.

4.2.3 Functional Requirements

LL-FR-INST-001: The system shall allow authorized users to create hospital, blood-centre, and organ-centre profiles.

LL-FR-INST-002: The system shall maintain institution name, type, contact, address, and location information.

LL-FR-INST-003: The system shall allow authorized institution users to update permitted profile information.

LL-FR-INST-004: The system shall maintain hospitals, blood centres, and organ centres as separate entities.

LL-FR-INST-005: The system shall allow an authorized blood centre to be optionally associated with a hospital or institution.

LL-FR-INST-006: The system shall allow an authorized organ centre to be optionally associated with a hospital or institution.

LL-FR-INST-007: The system shall display associated institutions or centres where the relationship is recorded and authorized for viewing.

LL-FR-INST-008: The system shall validate required institution information before saving a profile.

LL-FR-INST-009: The system shall maintain the operational status of participating institutions where applicable.

LL-FR-INST-010: The system shall make eligible institution information available to map-based discovery and coordination services.

LL-FR-INST-011: The system shall restrict institution-profile modification to authorized users.

LL-FR-INST-012: The system shall prevent unauthorized users from modifying institutional records.

LL-FR-INST-013: The system shall maintain records of significant institutional profile changes.

4.3 Blood Inventory Management

4.3.1 Description and Priority

This feature enables authorized blood centres and applicable institutions to maintain blood-resource availability and status. Inventory information serves as the operational source for blood-request coordination.

Priority: High

4.3.2 Stimulus/Response Sequences

- An authorized institution adds or updates a blood inventory record → the system validates the blood group, component, quantity, and other required information → the inventory is stored with its current status and timestamp → eligible coordination services can use the updated availability.
- When inventory is reserved, issued, expired, or otherwise changed → the system updates the available quantity/status and records the change.

4.3.3 Functional Requirements

LL-FR-INV-001: The system shall allow authorized institutions to create blood inventory records.

LL-FR-INV-002: Each inventory record shall contain the required blood group, component/type, quantity, status, and relevant date information.

LL-FR-INV-003: The system shall validate inventory information before storing it.

LL-FR-INV-004: The system shall prevent inventory quantities from becoming negative.

LL-FR-INV-005: The system shall allow authorized users to update inventory quantities and statuses.

LL-FR-INV-006: The system shall maintain timestamps for inventory updates.

LL-FR-INV-007: The system shall support inventory statuses such as available, reserved, unavailable, or expired where applicable.

LL-FR-INV-008: The system shall update available quantities when inventory is reserved or consumed through an authorized workflow.

LL-FR-INV-009: The system shall identify inventory information that is stale, unavailable, or otherwise not confirmed as current.

LL-FR-INV-010: The system shall provide eligible inventory information to blood-request coordination services.

LL-FR-INV-011: The system shall restrict inventory modification to authorized institution users.

LL-FR-INV-012: The system shall maintain records of significant inventory changes.

4.4 Blood Request and Matching

4.4.1 Description and Priority

This feature enables authorized users and institutions to create blood requirements, identify potential available sources, and coordinate fulfilment based on operational availability, location, and request conditions.

Priority: High

4.4.2 Stimulus/Response Sequences

- A user or institution creates a blood requirement → the system validates the request → searches eligible institutional inventory and applicable opted-in donor sources → evaluates availability and location → presents potential matches → authorized users coordinate acceptance or rejection → the request status is updated.

4.4.3 Functional Requirements

LL-FR-BREQ-001: The system shall allow authorized users and institutions to create blood requests.

LL-FR-BREQ-002: A blood request shall record required blood group, component/type, quantity, location, priority, and relevant request information.

LL-FR-BREQ-003: The system shall validate required blood-request information before submission.

LL-FR-BREQ-004: The system shall assign an identifiable status to each blood request.

LL-FR-BREQ-005: The system shall search eligible institutional inventory based on request requirements.

LL-FR-BREQ-006: The system shall support location/radius criteria when searching for potential sources.

LL-FR-BREQ-007: The system shall consider current availability and relevant freshness information when presenting institutional matches.

LL-FR-BREQ-008: The system shall support eligible opted-in donor sources where the applicable workflow permits donor coordination.

LL-FR-BREQ-009: The system shall present potential matches to authorized users without representing them as guaranteed fulfillment.

LL-FR-BREQ-010: The system shall support acceptance, rejection, reservation, fulfillment, and closure of applicable request workflows.

LL-FR-BREQ-011: The system shall update request status as coordination progresses.

LL-FR-BREQ-012: The system shall notify relevant participants when significant request-status changes occur.

LL-FR-BREQ-013: The system shall maintain a record of significant blood-request and matching activities.

4.5 Map-Based Discovery

4.5.1 Description and Priority

This feature provides location-based discovery of hospitals, blood centres, and organ centres to support resource identification and coordination.

Priority: High

4.5.2 Stimulus/Response Sequences

- A user selects a location or permits location access → the system retrieves eligible facility information → nearby hospitals, blood centres, and organ centres are identified according to the selected search criteria → results are displayed on a map and/or list.

4.5.3 Functional Requirements

LL-FR-MAP-001: The system shall provide map-based discovery of participating healthcare facilities.

LL-FR-MAP-002: The system shall display eligible hospitals, blood centres, and organ centres as separate facility types.

LL-FR-MAP-003: The system shall support location-based facility searches.

LL-FR-MAP-004: The system shall support radius or proximity-based searches where applicable.

LL-FR-MAP-005: The system shall display relevant facility information for selected results.

LL-FR-MAP-006: The system shall provide facility location information through map and/or list views.

LL-FR-MAP-007: The system shall support filtering results by facility type.

LL-FR-MAP-008: The system shall support filtering based on relevant availability or service information where available.

LL-FR-MAP-009: The system shall use current authorized location and institution data available to the platform.

LL-FR-MAP-010: The system shall indicate when required location or facility information is unavailable or outdated.

LL-FR-MAP-011: The system shall respect authorization rules when displaying restricted facility information.

LL-FR-MAP-012: The system shall provide appropriate feedback when no facilities satisfy the selected criteria.

4.6 Emergency Coordination and Notifications

4.6.1 Description and Priority

This feature supports time-sensitive healthcare-resource coordination by prioritizing emergency requirements, identifying nearby potential resources, broadcasting requests, and tracking responses.

Priority: High

4.6.2 Stimulus/Response Sequences

- An authorized user creates an emergency requirement → the system validates and marks the request with emergency priority → nearby eligible resources are identified → emergency notifications or broadcasts are issued to relevant participants → responses are received and tracked → the request status is updated as coordination progresses.

4.6.3 Functional Requirements

LL-FR-EMG-001: The system shall allow authorized users to create emergency coordination requests.

LL-FR-EMG-002: The system shall distinguish emergency requests from normal and urgent requests.

LL-FR-EMG-003: Emergency requests shall record the required resource, quantity, location, priority, and relevant contact information.

LL-FR-EMG-004: The system shall validate emergency-request information before initiating coordination.

LL-FR-EMG-005: The system shall identify eligible nearby institutions or other permitted sources.

LL-FR-EMG-006: The system shall support emergency broadcasting to relevant authorized participants.

LL-FR-EMG-007: The system shall provide notifications for emergency requests and significant responses.

LL-FR-EMG-008: The system shall record responses received from participating sources.

LL-FR-EMG-009: The system shall maintain emergency request status throughout coordination.

LL-FR-EMG-010: The system shall provide real-time or near-real-time updates where the required communication services are available.

LL-FR-EMG-011: The system shall maintain an auditable record of emergency coordination activities.

LL-FR-EMG-012: Emergency coordination shall not autonomously determine clinical treatment, medical eligibility, or other professional decisions.

4.7 Organ Coordination

4.7.1 Description and Priority

This feature provides structured support for organ-donation and transplantation coordination, including donor and organ records, recipient requirements, matching and offer workflows, procurement, transport tracking, and related records.

Priority: High

4.7.2 Stimulus/Response Sequences

- An authorized centre records donor and organ information → eligible recipient requirements are retrieved → potential matches are identified according to configured criteria → an offer or coordination workflow is initiated → authorized participants respond → procurement and transport statuses are updated → the workflow is completed and recorded.

4.7.3 Functional Requirements

LL-FR-ORG-001: The system shall allow authorized users to create and maintain donor records.

LL-FR-ORG-002: The system shall record applicable donor consent and authorization information.

LL-FR-ORG-003: The system shall allow authorized users to create and maintain organ records.

LL-FR-ORG-004: The system shall record relevant organ availability and workflow status.

LL-FR-ORG-005: The system shall allow authorized users to create and maintain recipient requirements.

LL-FR-ORG-006: Recipient requirements shall contain applicable structured information required for coordination.

LL-FR-ORG-007: The system shall identify potential donor-recipient matches using configured and authorized criteria.

LL-FR-ORG-008: The system shall present matching results as coordination information rather than autonomous clinical eligibility decisions.

LL-FR-ORG-009: The system shall support creation and tracking of organ offers.

LL-FR-ORG-010: The system shall record offer responses and applicable status changes.

LL-FR-ORG-011: The system shall support procurement workflow tracking.

LL-FR-ORG-012: The system shall record relevant procurement status and timestamps.

LL-FR-ORG-013: The system shall support transport workflow tracking for applicable organs.

LL-FR-ORG-014: The system shall maintain transport-related statuses and timestamps.

LL-FR-ORG-015: The system shall support notifications for relevant organ-coordination events.

LL-FR-ORG-016: The system shall maintain records of significant donor, recipient, offer, procurement, and transport activities.

LL-FR-ORG-017: The system shall restrict sensitive organ-coordination information according to authorization rules.

LL-FR-ORG-018: The system shall support post-coordination and outcome records where applicable.

LL-FR-ORG-019: The system shall not autonomously determine final medical eligibility or organ allocation.

LL-FR-ORG-020: The system shall support auditability of significant organ-coordination actions.

4.8 Real-Time Workflow Tracking

4.8.1 Description and Priority

This feature provides timely status updates and notifications for relevant coordination activities across LifeLink.

Priority: Medium

4.8.2 Stimulus/Response Sequences

- A relevant system event occurs → the backend updates the corresponding state → subscribed or authorized participants receive a real-time update and/or notification → the event is reflected in the applicable interface.

4.8.3 Functional Requirements

LL-FR-RTN-001: The system shall provide real-time updates for applicable workflow events.

LL-FR-RTN-002: The system shall deliver notifications to authorized and relevant recipients.

LL-FR-RTN-003: The system shall support notifications for request-status changes.

LL-FR-RTN-004: The system shall support notifications for emergency coordination events.

LL-FR-RTN-005: The system shall support notifications for relevant organ-coordination events.

LL-FR-RTN-006: The system shall provide appropriate notification status such as unread or read where applicable.

LL-FR-RTN-007: The system shall prevent unauthorized users from receiving restricted notifications.

LL-FR-RTN-008: The system shall maintain relevant notification records.

LL-FR-RTN-009: The system shall provide appropriate feedback when real-time delivery is temporarily unavailable.

LL-FR-RTN-010: The system shall synchronize applicable status changes with the operational data source.

LL-FR-RTN-011: The system shall prevent duplicate or inconsistent state updates resulting from repeated event processing.

4.9 RAG-Based Intelligent Assistance

4.9.1 Description and Priority

This feature provides context-aware intelligent assistance using retrieval-augmented generation. The RAG layer retrieves relevant information from approved knowledge sources and provides grounded assistance without replacing the operational database or professional decision-making.

Priority: Medium

4.9.2 Stimulus/Response Sequences

- A user submits a natural-language query → the system interprets the query → relevant approved knowledge is retrieved through the embedding and vector-search layer → applicable system context is obtained through authorized backend services where required → the LLM generates a grounded response → the response is presented to the user with appropriate limitations.

4.9.3 Functional Requirements

LL-FR-RAG-001: The system shall provide a natural-language interface for intelligent assistance.

LL-FR-RAG-002: The system shall process user queries and identify relevant information needs.

LL-FR-RAG-003: The system shall generate embeddings for supported knowledge content using the configured embedding service.

LL-FR-RAG-004: The system shall retrieve relevant information from approved knowledge sources.

LL-FR-RAG-005: The system shall use retrieved context when generating responses to applicable queries.

LL-FR-RAG-006: The system shall use the configured LLM for natural-language response generation.

LL-FR-RAG-007: The system shall distinguish knowledge-based information from live operational information.

LL-FR-RAG-008: Live operational information shall be obtained through authorized application/backend services rather than treated as vector-store data.

LL-FR-RAG-009: The system shall enforce authorization when AI assistance accesses protected operational context.

LL-FR-RAG-010: The system shall provide appropriate responses when relevant knowledge cannot be retrieved.

LL-FR-RAG-011: The system shall avoid presenting generated responses as authoritative medical or regulatory decisions.

LL-FR-RAG-012: The system shall not autonomously modify protected operational records through the LLM.

LL-FR-RAG-013: The system shall support maintenance and updating of approved knowledge sources.

LL-FR-RAG-014: The system shall maintain appropriate records of significant AI-assisted interactions where required for audit or system operation.

4.10 Predictive Analytics

4.10.1 Description and Priority

This feature provides predictive analysis for blood-resource planning, including demand forecasting and shortage-risk classification based on available historical and operational data.

Priority: Medium

4.10.2 Stimulus/Response Sequences

- Authorized data is provided to the predictive analytics service → the data is validated and processed → the configured model generates demand or shortage-risk outputs → results are returned to the application → authorized users view the outputs for planning and coordination.

4.10.3 Functional Requirements

LL-FR-PRED-001: The system shall provide a predictive analytics service for blood-resource planning.

LL-FR-PRED-002: The system shall use authorized historical and applicable operational data for predictive analysis.

LL-FR-PRED-003: The system shall validate input data before performing predictive analysis.

LL-FR-PRED-004: The system shall support blood-demand forecasting.

LL-FR-PRED-005: The system shall support shortage-risk classification.

LL-FR-PRED-006: The system shall provide predictive outputs for relevant blood groups, components, locations, or periods where sufficient data is available.

LL-FR-PRED-007: The system shall identify when available data is insufficient for reliable analysis.

LL-FR-PRED-008: The system shall store or return predictive results with relevant timestamps and analysis context.

LL-FR-PRED-009: The system shall make predictive outputs available to authorized users for planning.

LL-FR-PRED-010: The system shall distinguish predictive results from confirmed inventory information.

LL-FR-PRED-011: The system shall not automatically modify operational inventory based solely on predictive results.

LL-FR-PRED-012: The system shall not use predictive outputs as autonomous clinical or medical decisions.

LL-FR-PRED-013: The system shall support updating or retraining predictive models when authorized data and model procedures permit.

LL-FR-PRED-014: The system shall provide appropriate indication of prediction status or limitations where applicable.

LL-FR-PRED-015: The system shall maintain relevant records of predictive analyses for traceability.

4.11 Document/File Management

4.11.1 Description and Priority

This feature manages documents and supporting records associated with users, institutions, requests, and coordination workflows while applying appropriate access controls.

Priority: Medium

4.11.2 Stimulus/Response Sequences

- An authorized user uploads or creates a document/record → the system validates the file or information → stores it with associated metadata and access permissions → authorized users can retrieve or update it according to their permissions.

4.11.3 Functional Requirements

LL-FR-DOC-001: The system shall allow authorized users to upload permitted documents.

LL-FR-DOC-002: The system shall associate documents with relevant users, institutions, requests, or workflows.

LL-FR-DOC-003: The system shall maintain metadata required to identify and manage stored documents.

LL-FR-DOC-004: The system shall validate supported document types and applicable size restrictions.

LL-FR-DOC-005: The system shall store documents using the configured storage service.

LL-FR-DOC-006: The system shall allow authorized users to retrieve permitted documents.

LL-FR-DOC-007: The system shall restrict document access according to authorization rules.

LL-FR-DOC-008: The system shall support updating or replacing permitted documents.

LL-FR-DOC-009: The system shall maintain relevant document status and timestamps.

LL-FR-DOC-010: The system shall support deletion or archival of documents according to authorization and retention rules.

LL-FR-DOC-011: The system shall maintain operational records associated with important coordination workflows.

LL-FR-DOC-012: The system shall preserve relationships between records and their associated workflows or entities.

LL-FR-DOC-013: The system shall prevent unauthorized modification or deletion of protected records.

LL-FR-DOC-014: The system shall protect sensitive documents and records during storage and transfer.

LL-FR-DOC-015: The system shall maintain appropriate audit information for significant document and record operations.

4.12 Audit and Administration

4.12.1 Description and Priority

This feature provides system-level administration, access management, monitoring, and audit capabilities to support controlled operation and traceability.

Priority: High

4.12.2 Stimulus/Response Sequences

- An authorized administrator accesses an administrative function → the system verifies administrative privileges → the requested account, configuration, monitoring, or audit operation is performed → the relevant result and activity are recorded.

4.12.3 Functional Requirements

LL-FR-ADM-001: The system shall provide an administrative interface for authorized administrators.

LL-FR-ADM-002: The system shall allow authorized administrators to manage user accounts.

LL-FR-ADM-003: The system shall allow authorized administrators to manage applicable roles and access permissions.

LL-FR-ADM-004: The system shall allow administrators to activate, deactivate, or restrict accounts according to authorization.

LL-FR-ADM-005: The system shall provide administrative visibility into relevant institutions and system resources.

LL-FR-ADM-006: The system shall provide monitoring information for important system and workflow activities.

LL-FR-ADM-007: The system shall record significant administrative actions.

LL-FR-ADM-008: The system shall record significant authentication and authorization events where required.

LL-FR-ADM-009: The system shall record significant changes to operational records where auditability is required.

LL-FR-ADM-010: The system shall provide authorized administrators with access to audit records.

LL-FR-ADM-011: The system shall restrict audit records from unauthorized modification or deletion.

LL-FR-ADM-012: The system shall support filtering or searching of audit records using relevant criteria.

LL-FR-ADM-013: The system shall support administrative monitoring of important request and coordination statuses.

LL-FR-ADM-014: The system shall enforce authorization checks for all administrative operations.

LL-FR-ADM-015: The system shall prevent administrators from accessing information beyond their assigned authorization scope where applicable.

LL-FR-ADM-016: The system shall maintain timestamps and relevant actor information for auditable activities.

LL-FR-ADM-017: The system shall provide appropriate feedback when an administrative operation fails or is unauthorized.

LL-FR-ADM-018: The system shall support auditability of significant system configuration and access-control changes.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

LifeLink shall provide responsive operation for normal user interactions and coordination workflows under the expected system load. Performance requirements apply to application responses, database operations, real-time communication, search, and intelligence services.

PERF-001: The system shall provide timely responses for normal user interactions and commonly used application functions.

PERF-002: The system shall process database operations efficiently enough to support concurrent users and institutional workflows within the expected system capacity.

PERF-003: The system shall support location-based searches without unnecessary processing or retrieval of unrelated records.

PERF-004: The system shall provide blood inventory and request information using the most recent authorized operational data available.

PERF-005: The system shall support real-time updates and notifications with minimal avoidable delay when the required services are available.

PERF-006: The system shall process emergency coordination requests with priority over non-critical background operations where applicable.

PERF-007: The system shall avoid unnecessary retrieval of large document or knowledge collections during RAG-based queries by using relevant retrieval mechanisms.

PERF-008: The predictive analytics service shall process supported datasets within a reasonable time for the intended planning workflow.

PERF-009: The system shall remain responsive when external AI, mapping, or other dependent services experience temporary delays, where alternative application functionality remains available.

PERF-010: Performance-sensitive operations shall be designed to support future increases in users, institutions, requests, and operational records within the intended system scope.

5.2 Safety Requirements

LifeLink shall incorporate safeguards to reduce risks arising from incorrect, outdated, inconsistent, or unavailable information. AI and predictive components shall support coordination and decision-making without independently performing clinical or regulatory decisions.

SAFE-001: The system shall identify or appropriately handle stale, unavailable, or unverified operational information.

SAFE-002: The system shall validate critical inputs and prevent invalid operational states such as negative inventory quantities.

SAFE-003: The system shall preserve valid persisted workflow states during temporary connectivity or real-time communication failures.

SAFE-004: The system shall clearly distinguish normal, urgent, and emergency coordination requests.

SAFE-005: AI-generated information shall not be treated as the authoritative source for live operational data unless verified through authorized system data.

SAFE-006: The system shall not use AI to independently diagnose conditions, determine medical eligibility, determine transfusion compatibility, prescribe treatment, or make final organ-allocation decisions.

SAFE-007: Predictive analytics shall be used as decision support and shall not independently trigger clinical or medical decisions.

SAFE-008: The system shall provide appropriate indications when required data or dependent services are unavailable or when an AI-generated response cannot be adequately grounded.

5.3 Security Requirements

LifeLink shall protect user accounts, institutional information, coordination records, documents, and other sensitive information through authentication, authorization, controlled data access, secure communication, and auditability.

SEC-001: The system shall require authentication before granting access to protected functionality.

SEC-002: The system shall enforce role-based authorization for protected resources and operations.

SEC-003: The system shall restrict access to sensitive information according to user and institutional authorization.

SEC-004: The system shall validate and appropriately sanitize user-provided input.

SEC-005: The system shall protect sensitive information during transmission and storage using appropriate security mechanisms.

SEC-006: The system shall record significant security-sensitive and administrative activities for audit purposes.

SEC-007: The system shall protect audit records from unauthorized modification or deletion.

SEC-008: Real-time events, notifications, and AI/RAG interactions shall respect applicable authorization rules.

SEC-009: The system shall prevent unauthorized access to administrative and other protected resources, including through direct backend requests.

SEC-010: The system shall minimize exposure of sensitive information in logs, notifications, error messages, and user interfaces.

5.4 Software Quality Attributes

LifeLink shall be designed to provide usability, reliability, maintainability, adaptability, and consistency across its major components.

QUAL-001: The system shall provide interfaces appropriate to the responsibilities of each user class.

QUAL-002: The system shall provide clear feedback for successful, unsuccessful, and pending operations.

QUAL-003: The system shall maintain consistent navigation, terminology, and interaction patterns across major modules.

QUAL-004: The system shall maintain consistency between related operational records and workflow states.

QUAL-005: The system shall provide appropriate recovery or error handling when dependent services become temporarily unavailable.

QUAL-006: The system shall maintain logs and audit information required for troubleshooting and traceability.

QUAL-007: The system shall use modular components so that individual services or features can be maintained or extended without unnecessary changes to unrelated components.

QUAL-008: Operational data, RAG knowledge data, predictive outputs, and application logic shall remain appropriately separated.

QUAL-009: The system shall allow approved knowledge sources used by RAG to be updated without requiring changes to unrelated operational data.

QUAL-010: The system shall provide meaningful error messages without unnecessarily exposing internal implementation or sensitive information.

QUAL-011: The system shall maintain compatibility with supported modern web browsers and device types.

QUAL-012: The system shall maintain sufficient documentation of source code, configuration, dependencies, and deployment procedures to support system maintenance.

QUAL-013: The system architecture shall support future extension of healthcare coordination, intelligence, and institutional capabilities within the defined scope.

5.5 Business Rules

The following business rules define important operational and organizational constraints that govern LifeLink.

- **BR-001:** Access to protected functionality and information shall be determined by the user's assigned role and authorization.
- **BR-002:** Hospitals, blood centres, and organ centres shall remain separate system entities, with optional institutional associations where applicable.
- **BR-003:** Blood inventory shall be maintained by authorized institutional users and shall reflect the availability status recorded by the responsible institution.
- **BR-004:** Blood matching shall use operational availability, request requirements, location, and applicable coordination criteria; it shall not independently determine clinical compatibility.
- **BR-005:** Emergency workflows shall support prioritized coordination, notification, and response tracking within the capabilities of the system.
- **BR-006:** Organ coordination shall support structured donor, recipient, matching, offer, procurement, and transport workflows without autonomously determining final medical eligibility or organ allocation.
- **BR-007:** The structured operational database shall remain the source of truth for live operational information; the RAG knowledge store shall not be used as the source of live inventory or request status.
- **BR-008:** AI-generated outputs shall operate within applicable authorization and business rules and shall not independently modify protected operational records.
- **BR-009:** Significant operational, administrative, and security-sensitive activities shall be traceable through appropriate audit records.

- **BR-010:** LifeLink shall function as a coordination and decision-support platform and shall not replace professional clinical judgment, regulated healthcare processes, or official organ-allocation authorities.

6. Other Requirements

The following requirements address areas not covered in Sections 3–5.

6.1 Database Requirements

The operational database shall use the core entities defined in Section 2, related through foreign keys and indexed on commonly searched fields such as blood group and location.

6.2 Document/File Storage Requirements

Supporting documents and files shall be stored in object storage, with corresponding metadata and references maintained in PostgreSQL. The retention policy for documents, audit logs, and inactive accounts is tracked in Appendix C.

6.3 Localization and Internationalization

LifeLink shall provide an English-language interface in the current version. Support for additional languages is outside the scope of this SRS.

6.4 Data and Evaluation Requirements

Testing and predictive-analytics evaluation shall use synthetic or anonymised data. Results shall be treated as system-level indicators and not as clinically validated outcomes.

6.5 Regulatory and Ethical Requirements

LifeLink shall be used as a coordination and decision-support platform and shall not be treated as a substitute for clinical judgment, regulatory processes, or required clinical authorization.

Appendix A: Glossary

Term	Definition
RBAC	Role-Based Access Control — an authorization model that assigns system permissions according to a user's role.
RAG	Retrieval-Augmented Generation — retrieval of relevant knowledge from a store and its use as context for language-model responses.
LLM	Large Language Model — the language-generation model used by the AI Assistant.
Vector Store	A data store containing document chunks and embeddings for semantic retrieval.
Embedding	A numerical vector representation used to perform semantic similarity search.
Geospatial Discovery	Location-based search and distance/radius filtering of registered facilities, distinct from continuous GPS tracking.
Blood Inventory	Recorded blood stock held by a Blood Bank or Hospital, tracked by group, component, quantity, and status.
Blood Component	A specific blood product, such as red blood cells, plasma, or platelets.
Shortage Risk	A classification indicating the likelihood of insufficient supply of a blood group or component.
Organ Centre	An institution responsible for organ-related records, coordination, and associated workflow activities.
Recipient/Candidate	An individual represented in the system as requiring a blood or organ resource, as applicable.
Donor	An individual registered as available or consenting to contribute blood or organs, as applicable.
Request	A structured record of a blood or organ requirement tracked through a defined workflow.
Match	A candidate pairing identified by the system between a request and available inventory, donor, or organ record using defined rules.
Offer	A structured proposal arising from a match and tracked through review and acceptance or decline.
Workflow	The defined sequence of states through which a request, match, or offer progresses.
Audit Log	A traceable record of significant state-changing actions performed in the system.
Emergency Broadcast	A structured urgent notification sent to relevant nearby institutions and eligible opted-in users.
Socket.IO	The real-time communication library used to deliver system events to connected clients.
Pub/Sub	a messaging pattern used with Redis to distribute events between system components.

Appendix B: Analysis Models

The following analysis models are relevant to LifeLink:

- Layered Architecture Diagram — Users/Actors → Frontend → Application Layer → Intelligence Layer → Data Layer → Deployment & Delivery. (Prepared)
- Use-Case Diagram — Interactions of Hospital, Blood Bank, Organ Centre, User, and Administrator. (Planned)
- Context Diagram — LifeLink boundary and its relationship with external systems. (Planned)
- Entity–Relationship Diagram — Core data entities and their relationships. (Planned)
- Blood-Request State Diagram — Request lifecycle from creation to closure; final states are tracked in Appendix C. (Planned)
- Sequence Diagram — End-to-end blood-request coordination flow. (Planned)
- RAG Workflow Diagram — Query embedding, retrieval, context assembly, and response generation. (Planned)
- Organ-Coordination Workflow Diagram — Organ request, matching, offer, and response flow. (Planned)

Appendix C: To Be Determined List

The following items are genuinely undecided at the time of this SRS version and are tracked here rather than assumed. Once decided, each item should be resolved throughout the document and removed from this list.

#	TBD Item	Related Section(s)
TBD-1	Exact machine-learning algorithm(s) for blood-demand forecasting and shortage-risk classification.	4.10, 5.4
TBD-2	Exact vector-database technology for the RAG knowledge store.	2.1, 3.3, 4.9
TBD-3	Exact notification provider(s) for e-mail/SMS delivery.	3.4, 4.6
TBD-4	Final deployment hosting environment.	2.4, 2.6
TBD-5	Final database schema details beyond the conceptual entity list.	6.1, Appendix A
TBD-6	Final request-status and organ-workflow state names (current names are illustrative/conceptual).	4.4, 4.7
TBD-7	Quantitative performance targets beyond the demo-scale estimates in Section 5.1, once real load-testing is performed.	5.1
TBD-8	Final document/file and audit-log retention policy.	6.2
TBD-9	Final UI screen-level specifications (wireframes/mockups) beyond the feature-level description in Section 3.1.	2.6, 3.1
TBD-10	Whether and how formal end-user documentation (manuals) will be produced beyond in-application guidance and the AI Assistant.	2.6